

Death

Player Death

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User starts or continues a run		User is in game, not in the main menu	
User dies		A death animation plays, a stats screen appears	
User presses space bar		User is returned to the main screen	

Enemy Death

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User starts or continues a run		User is in game, not in the main menu	
User kills enemy		Enemy despawns	

Graphics/Animation

Graphics load in

Step	Data/Special Condition	Expectation	Bug
User launches the game			

User looks at player character		Should see correctly loaded in textured character	
User looks at walls		Should see correctly loaded in textured walls	
User looks at enemies		Should see correctly loaded in textured enemies	
User looks at floor		Should see correctly loaded in textured floor	
User looks at bullets		Should see correctly loaded in textured bullets	

Melee Animation - Enemy

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User walks into either enemy		User should see melee arm swinging animation	

Movement Animation

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User walks		User should see walking animation	
User triggers both enemy animations		User should see walking animations for both	

NPC Animation

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User encounters an NPC		User should see an idle animation	

Movement

Basic Movement

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User clicks 'W' key		Character moves up. Should have no delay and be responsive.	
User clicks 'A' key		Character moves left. Should have no delay and be responsive.	
User clicks 'S' key		Character moves down. Should have no delay and be responsive.	
User clicks 'D' key		Character moves right. Should have no delay and be responsive.	
User clicks 'W' and 'D' key together		Character moves diagonally NE	
User clicks 'W' and 'A' key together		Character moves diagonally NW	
User clicks 'S' and 'D' key together		Character moves diagonally SE	

User clicks 'S' and 'A' key together		Character moves diagonally SW	
User looks at the way the character and enemies move		Interpolation should be present on movement	

Opposite Buttons Pressed

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User clicks 'W' and 'S' key together		Character does not move	
User clicks 'A' and 'D' key together		Character does not move	

Shooting

Basic Shooting

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User clicks 'LMB' on window		A bullet is fired in the direction of the cursor position	

User moves cursor somewhere else on the screen and clicks 'LMB'		A bullet is fired in the direction of the new cursor position	
User holds 'LMB'		Bullets are fired at a steady rate in the direction of the cursor position	
User holds 'LMB' and drags cursor around		Bullets are fired at a steady rate in the direction of the cursor as it moves	
Certain amount of time passes after bullet is fired		Bullet should despawn after a certain amount of time	

Shooting While Moving

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User clicks 'LMB' on window		A bullet is fired in the direction of the cursor position	
User moves character and then clicks 'LMB'		A bullet is fired in the direction of the cursor position and the character continues to move. Bullet stays on straight path	
User stops			
User moves character and then clicks 'LMB'		Bullets are fired at a steady rate in the direction of the cursor as it moves	
Repeat above but with holding 'LMB'			

Collision

Wall Collision

Step	Data/Special Condition	Expectation	Bug
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User launches the game			
User moves character into wall and continues to walk into it		The character should remain still next to the wall. The character should not clip or jitter through wall.	
User moves away from wall		Character is not stuck in wall	

Corner Wall Collision

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User moves character into corner and continues to walk into it		The character should remain still next to the wall. The character should not clip or jitter through the wall.	
User moves away from wall		Character is not stuck in wall	

Floor Object Collision

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User moves character into floor object and continues to walk into it	Applies to enemies as well	The character should remain still next to the object. The character should not clip or jitter through objects.	

User moves away from object		Character is not stuck in object	
Repeat by approaching at different angles			

Wall-Bullet Collision

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User shoots bullet at wall		The bullet is destroyed at an appropriate position, without clipping or jittering. Mesh of the projectile should be used	

Enemy-Bullet Collision

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User shoots bullet at enemy	Enemies in M2 should only take 5 hits to kill	The bullet is destroyed at an appropriate position, without clipping or jittering. And the appropriate amount of damage is applied. The player should also see particles fly out of the enemy with air friction. Mesh of the projectile should be used	

Player-Enemy Collision

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User collides with enemy		The appropriate amount of damage is applied to the player and enemy. The player should also see particles fly out of the enemy and themselves with air friction.	
User tries to collide with enemy immediately after		The player should also have 3s (subject to change) of invincibility and shouldn't take damage	
Invincibility timer runs out			
User collides with enemy again		The appropriate amount of damage is applied to the player and enemy. The player should also see particles fly out of the enemy and themselves with air friction.	

Player-Enemy Collision With 1 HP Left On player

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User collides with enemy while player only having one health left	Enemies have 5 HP, subject to change	Player dies and the stats screen appears.	

Pausing

Pause Game Esc Key

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User taps the 'ESC' key		The game is paused. Pause menu appears with save, back, help, and quit options	
User taps the 'ESC' key again		Game resumes. Game is back on screen just as it was left when paused	

Pause Game Resume Button

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User taps the 'ESC' key		The game is paused. Pause menu appears with save, back, help, and quit options	
User clicks the Resume button		Game resumes. Game is back on screen just as it was left when paused	

Pause Game Main Menu Button

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User taps the 'ESC' key		The game is paused. Pause menu appears with save, back, help, and quit options	
User clicks the Main Menu button		User is sent back to the main menu	

Pause Game Help Button

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User taps the 'ESC' key		The game is paused. Pause menu appears with save, back, help, and quit options	
User clicks the Help button		User sees the help screen, with a button to return to the pause screen.	
User clicks the Back button		User is returned to the pause screen.	

Interaction

Interact With Item

Step	Data/Special Condition	Expectation	Bug
User launches the game		Player spawned in a room with enemies	
User kills enemy who drops an item		An item appears on the ground at where the enemy is killed	

User moves on to an item			
User taps the 'E' key		The item disappears on the ground and appears in one of the item slots (right to left)	

Interact With NPC

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User goes to a room with NPC			
User moves near to the NPC			
User taps the 'E' key		Text box appears on the screen	
User keeps taps the 'E' Key		Text box new new dialogue appears in the text box until the conversation is over and the text box closes	

Health

Health Display

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User starts or continues a run			
User sees health in game UI		Should see a battery bar with green segments corresponding to the total hitpoints	

User takes damage			
User looks at health in game UI		Should 1 less health segment.	

UI

UI Appears

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User looks at top of screen	Values subject to change	Should see a battery bar with green segments corresponding to current health, a number for SCRP (scrap) and a number for LVL corresponding with current stage/level. They should also see at least 2 slots for items, and up to 8	
UI should update accordingly		Damage/health items increment the health value. Picking up items should remove the item from the ground and have the item show up in one of the item slots on top of the screen	

Pathfinding

Basic Enemy Pathfinding

Step	Data/Special Condition	Expectation	Bug
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User launches the game			
User observes enemies		Enemies move towards player	
User starts moving		Enemies continue to move towards player	
User stops moving		Ranged enemies continue to move towards player until reached a certain distance Melee enemies continue to move towards player	
User starts moving again		Enemies continue to move towards player	

FPS Counter

FPS Counter

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User looks at window header		The user should see a FPS counter that updates as they play the game which doesn't drop or increase in large amounts, may stay stable at a single value	

Items

Item Drops

Step	Data/Special	Expectation	Bug
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	Condition		
User launches the game			
User kills enemy	15% chance to drop item for now (M2)	Enemy drops an item where they died. Could be one of 5 items all with equal chance to drop (20% each)	
User observes item		Item texture is clearly visible and not flickering	
Repeat until every item is received			

Item Pickup

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User presses 'E' when within range of an item		Item disappears off of the ground and the appropriate stats are applied to the player modifier. The item is added to inventory.	
User continues playing		Buff/debuffs are applied and change the gameplay. See buildItemSet() in world_init.cpp for specific stat buffs/debuffs.	
Repeat until every item is picked up			

Drop item ground from inventory

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User presses 'E' when within range of an item		Item disappears off of the ground and the appropriate stats are applied to the player modifier. The item is added to inventory.	

User presses number from 1 to 9 to drop item of the corresponding slot	There is a corresponding item in the item slot	The item at the corresponding slot is no longer in that slot, and on the ground as an interactable instead.	
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Item Pickup Full

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User collects 8 items			
User presses 'E' when within range of an item		Item remains on the ground and nothing is added to the inventory	
User continues playing		Buff/debuffs should remain the same. See buildItemSet() in world_init.cpp for specific stat buffs/debuffs.	
Repeat until every item is picked up			

Item Scrapping

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User presses 'X' when within range of an item		Item disappears off of the ground and the scrap amount of the item is added to the 'sc' counter on top of the ui	

Audio

Music system

Step	Data/Special Condition	Expectation	Bug
User launches the game		Post combat music starts playing	
User enters a room with enemy		Combat music starts playing	
User shoots a bullet		A player bullet sound effect is played	
Enemy shoots a bullet		A enemy bullet sound effect is played	
User got hit the an enemy bullet		A enemy bullet damage sound effect is played	
User got melee attacked		A melee attack sound effect is played	
User kills all enemies		Post combat music starts playing	

Main Menu

Main Menu

Step	Data/Special Condition	Expectation	Bug
User presses “run” button		A new game starts	
User presses “continue” button	there is a saved game	Load the saved game	
User presses “help” button		Show game tutorial	
User presses “upgrades” button		Nothing happens for now	

User presses “quit” button		the game terminates	
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Game Tutorial Menu

Step	Data/Special Condition	Expectation	Bug
User presses “back” button		The screen goes back to the main menu	

Shop

Purchase upgrade with scrap

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User clicks the upgrades button		Shop system show up	
User clicks desired upgrade	player have enough scrap to pay for the upgrade	Description and price of the upgrade show up	
User clicks “BUY” text		Upgrade is purchased, scrap price is paid	

Maxxed out upgrade

Step	Data/Special Condition	Expectation	Bug
User launches the game			

User clicks the upgrades button		Shop system show up	
User clicks desired upgrade	Upgrade is already at max level	Description and price of the upgrade show up, "BUY LVL" button is replaced with "MAX LVL"	

Bosses

Boss 1 - Mother of Dispersion

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User starts or continues a run			
User encounters the first boss		The boss itself appears at the top of the room, and spawns waves of enemies - 4 in total	
User defeats a wave of enemies		Next wave spawns	
User defeats last wave of enemies		Boss dies, 2 items drop, door to next stage opens	

Boss 2 - Mother of Recycling

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User starts or continues a run			
User encounters the second boss		The boss spawns in the center of the room, moves around, and drops floor objects (obstacles). It	

		also periodically disables the player's items, and shoots projectiles.	
User defeats the boss		Boss dies, 2 items drop, door to next stage opens	

Boss 3 - Mother of Propagation

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User starts or continues a run			
User encounters the final boss		The boss spawns near the top of the room. 4 'elite' enemies are also spawned, which shoot projectiles and move around the room. As each one dies, the remaining enemies' behaviour changes.	
User defeats the boss's enemies		The boss begins shooting projectiles	
User defeats the boss		The user wins the run, a stats screen appears.	

Reloadability

Pause save button

Step	Data/Special Condition	Expectation	Bug
User launches the game			

User starts a new game by clicking 'new run'			
User taps the 'ESC' key		The game is paused. Pause menu appears with back and quit options	
User presses the "Save" button		Game goes to main menu, and the game is saved	

'Continue Run' button

Step	Data/Special Condition	Expectation	Bug
User launches the game			
User clicks 'continue' button in main menu		The user is placed into their previously saved in-progress run.	